

PHYSICS NOTES

Motion and Measurement

Speed describes how quickly distance changes over time. Velocity also includes direction. Acceleration describes how quickly velocity changes.

Quantity	Symbol	Unit
Distance	s	meter
Time	t	second
Speed	v	meter/second
Acceleration	a	meter/second squared

Testing idea: extract only page 1 and confirm the table remains sharp and searchable.

MATHEMATICS NOTES

Functions and Graphs

A function maps each input to exactly one output. Linear functions form straight lines, while quadratic functions form parabolas.

Linear

$$f(x) = mx + b$$

Quadratic

$$f(x) = ax^2 + bx + c$$

Exponential

$$f(x) = a^x$$

Logarithmic

$$f(x) = \log_a(x)$$

Domain, range, intercepts, and rate of change are useful properties when comparing functions.

CHEMISTRY NOTES

Atomic Structure

Atoms contain protons, neutrons, and electrons. The atomic number equals the number of protons, while the mass number is the total of protons and neutrons.

Particle	Charge	Location
Proton	+1	Nucleus
Neutron	0	Nucleus
Electron	-1	Electron cloud

This final page can test split-by-range operations such as 1-2 and 3.